Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |

AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Research engineer specializing in **AI observability and LLM evaluation** - built AgentProbe, an open-source monitoring toolkit with trace enrichment, LLM-powered judges, agentic investigation pipelines, and a 5-metric eval engine designed to maintain human oversight before every AI deployment. 8+ years backend engineering at scale (1M+ records, 100K+ MAU, Python/AWS). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

AI Observability & Monitoring | LLM Evaluation & Judges | Trace Enrichment & Dashboards | Agentic Integrations | LLM-as-Judge Pattern | Human Oversight Systems | LangChain / LangGraph | RAG Pipelines / ChromaDB | Python / FastAPI | Large-Scale Data Pipelines | AWS / Docker | Open-Source Developer Tools

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **large-scale data processing frameworks** on AWS (S3, Glue, EC2) handling **1M+ records** with automated quality checks, Libsodium encryption, and zero security incidents - eliminating 80% of manual operational workload.
- Improved production query performance by **30%** by architecting PostgreSQL schemas and data access layers across 20+ projects - the analytical discipline behind connecting offline metrics to real system improvements.
- Built **20+ production modules and internal tools** (Python, TypeScript, PostgreSQL) prioritizing reliability, documentation, and UX across **120+ Agile sprints** for **100K+ monthly active users**.
- Cut microservices execution time by **50%** via Docker containerization; maintained **99.5% uptime** and sub-10 bug count for 6+ consecutive months via TDD/BDD quality gates.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted Python and database mentoring for 30+ undergraduates through sprint-style feedback cycles.

PROJECTS

AgentProbe: AI Observability & LLM Evaluation Toolkit OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure

Built to maintain **human oversight of AI pipelines before deployment**. Architecture: **RAG Tracer agent** enriches every run with per-query trace data (retrieval scores, generation quality, anomaly flags); **5-metric LLM-as-judge engine** (correctness, hallucination, retrieval, completeness, compliance) surfaces unexpected patterns across runs; **Auto Fix agent** autonomously investigates eval failures and acts on findings - rewrites prompts, re-runs eval, closes loop without human intervention per cycle; **React observability dashboard** with full trace visualization; **15-endpoint FastAPI backend**. Open-source developer tool: documented API, multi-provider support (Ollama, OpenAI, Anthropic, Groq), runs 100% locally. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP pipeline mapping natural language queries (EN/ES) to structured FHIR APIs via LLMs, BERT, embeddings. HIPAA-compliant RBAC, containerized. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive Modeling on 100K+ Records

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle on UK road accident data. **85%+ accuracy**; interactive dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Observability: LangChain, LangGraph, LLM Judges, LLM-as-Judge, Trace Enrichment, RAG, ChromaDB, Embeddings, Prompt Engineering, Ollama, OpenAI, Anthropic, Groq, FastAPI

ML Stack: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, spaCy, Hugging Face, Feature Engineering

Data & Cloud: AWS (S3, EC2, RDS, Glue), Azure, PostgreSQL, MySQL, NoSQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Testing & Quality: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum